



School of Information Technology, Engineering, Mathematics and Physics

CS241 – Software Design & Implementation Semester 2, 2025

Assignment 1 – Software Design Document

Learning Outcomes:

This assignment tests the following course learning outcomes and the associated CBOK Attributes:

- CLO 1 -Develop a software design document.
- CLO4- Demonstrate ethical and social consideration to software development.

Due Date: Sunday 31 August 2025 @ 11:55pm (Fiji Standard Time)

Submission: This assignment is to be done in groups (group of four students). Submit the entire project folder as a zip file via the Moodle Assignment 1 Drop box.

Weight: 25%

Weight: 15% for Software Design Document, 5% for presentation skills, and 5% for the proof of complex computing in your project.

***Note:** The attached marking rubric on pages 10-17 will be used to assess this assignment.

PART – 1: Software Design Document

Requirements:

1. Proof the M level of complex-computing of your project in presentation.
2. Develop a software design document for your allocated case study.
3. Apply acceptable software design tools and techniques to model physical models for both structured and OO approaches.
4. Develop a team's terms of reference (ToR).
5. Plan the implementation.

Software Design Document Template

- There is a software design document template attached on page 4.
- All teams will need to adopt this template.
- Teams can add more sections that may be relevant to the assigned project.

Team Terms of Reference (ToR)

- All teams must develop a team ToR that will govern the organization, operation, and code of conduct for their teams.
- The template for the team ToR is available under Assignment 1. Include a short bio for each team member and strategies for peer review.
- All teams are to demonstrate the ethical behavior in the formation of the required report.

Reference

- Reference all materials you use.
- APA referencing style to be used.

Final Document

You should place the following items in one PDF document:

- Cover page (Please show group number, and case study title clearly).
- Software Design Report Document.
- Mark Allocation Sheet.

Submission Guidelines

- Submit a soft copy of the file through Moodle using the Project Leaders account.
- The filename should read [group No].zip for example, 3.zip.
- Incorrect submissions will result in loss of marks or simply a mark of zero.

LATE SUBMISSION = ZERO (0)

Plagiarism

- No two teams should submit the same or similar assignment.

Queries

- Maintain contact through Moodle discussion forum named *Assignment Discussion Forum*.
- Feel free to consult the Lecturer/Course Coordinator during the online consultation hour or tutorial hours for queries.

Team Work

- You are required to form a team of 4 members for this assignment.
- You will have to form your teams and have them registered on Moodle by week 4 end.

- You will also have to indicate to the lecturer on the project case study that your team will be working on in the discussion that will be created on Moodle (Assignment Confirmed Topics).

Software Design Report Template

1. Project Title & Team details
2. Introduction & Purpose of this document. Describe its social significance.
3. System Capabilities and Scope (from the SRS document)
 - 3.1. Stakeholders
 - 3.2. Functional Requirements (describe conflicting requirements (if any))
 - 3.3. Non-Functional Requirements (describe conflicting requirements (if any))
4. Risk Analysis (related to Design)
5. Data Modeling: Entity Relationship Diagram
6. Application Architecture
 - 6.1. Structured design artifacts
 - 6.1.1. Structured Chart
 - 6.1.2. System Context Diagram/DFD Level 0 Diagram
 - 6.1.2.1. Primitive DFD diagram (Lowest level DFD) for a major module.
 - 6.2. Object-Oriented design artifacts
 - 6.2.1. Use-Case Diagram and description
 - 6.2.2. Domain Model Class diagram
 - 6.2.3. A major Sequence Diagram (s)
 - 6.2.4. Detailed/Design Class Diagram
7. User Interfaces
8. Implementation Plan
9. Definitions and Acronyms

Notes: How to create this document?

1. Introduction & Purpose of this document

Under this section introduce the problem domain of your given case study, the purpose of this design document and its intended users. Explain how organisational context such as type, size and staff composition of the project client organisation is applied to your design of the software solution. Identify any design standards that have been adopted/followed.

2. Entity Relationship Diagram

Create the entity-relationship diagram showing the database structure for your information system. Make sure the tables shown are in third normal form. (Using the event table as input, extract all possible nouns for which the system needs to store data about.)

3. Risk Analysis (related to Design)

Identify risks associated with the design of your proposed systems and mitigation strategies.

4. DFD Level 0 Diagram (Event-Partitioned System Model)

Create diagram 0 for the system. If the diagram is too complex then consider creating sub-systems.

5. Structure Chart

Design the interactions of all the modules of your system (or a major module) that will be required to implement the processes from DFD Level 0 Diagram.

6. Use Case Diagram

Create the use case diagram for your information system. Show all use cases of your system. Describe a major use-case.

7. Domain Model Class Diagram

Draw the entity class diagram for the entire system.

8. Detailed/Design Class Diagram

Draw the detailed class diagram showing the boundary, control, entity and data access classes used to implement the selected use case.

9. Sequence Diagrams

Create the sequence diagram required to implement the selected use case. Show all classes - boundary, control, entity and data access classes.

10. User Interfaces

For the selected use case or DFD process, create appropriate prototype user interfaces (forms/storyboards). You can use any IDE or MS Visio or wireframe tools to create the user interfaces. Alternatively, you can sketch the storyboards using any drawing/sketching tool.

11. Implementation Plan

Prepare in tabular format the intended plan of the implementation of your project for the second half of the semester. Show the components/sub systems to be developed and person assigned to implement that component. Show approximate timelines and resource requirements. The duration of this plan should cover a period from the week 8 to week 12.

Design Considerations:

1. **Scalability:** The system should handle a large number of students, teachers, and parents, especially during peak times like the end of a term.
2. **User Experience:** The user interface should be intuitive, responsive, and mobile-friendly, ensuring easy access for students, teachers, and parents.
3. **Data Integrity:** Measures should be in place to prevent data loss, corruption, and unauthorized access to sensitive information.
4. **Performance:** The system should provide real-time updates on class schedules, grades, and attendance, and respond quickly to user interactions.
5. **Integration:** The system might need to integrate with external systems, such as email notification services, grade management tools, and educational databases.
6. **Testing:** Thorough testing should ensure the system's functionality, security, and performance, including user acceptance testing by teachers, students, and parents.
7. **Documentation:** Comprehensive user guides and admin manuals should be available for students, teachers, parents, and administrators to ensure proper use of the system.

Note for Assignment:

- You can also capture other important requirements not specified above and design considerations in detail in your documentation.
- Be clear about the architecture you'll use and why.
- Apply different design techniques to explain the solution.
- The selected option to be implemented for assignment 2.

Assignment Case Studies

Your team is tasked to choose any one of the case studies for the assignment and requires you to evaluate and prepare feasibility study for their needs.

Option 1: Indigenous Knowledge System

Our Indigenous communities are enriched with rich traditional knowledge and skills that are passed down from generations to generation across the world. These knowledges can be categorised in various domains such as medicine, agriculture, environment management, cultural heritage, oral traditions, craftsmanship and our governance systems. However, as we move towards globalisation, modernisation and the low transmission of knowledge many of these are being impose risk of being lost.

To protect, document and disseminate Indigenous Knowledge while assuring its secure administration and ethical usage, an Indigenous Knowledge System must be developed. This system should function as a digital knowledge repository, including modern technologies to improve accessibility, usability and community ownership.

Currently, Indigenous communities and knowledge holders face several challenges:

- Lack of documentation leads to loss of critical knowledge.
- The younger generation has limited access to record traditional knowledge
- Risks of misappropriation and intellectual property theft.
- Disconnection between traditional and current scientific knowledge systems.
- Lack of centralised channels for information sharing among Indigenous groups.

The primary goal of this project is to design and implement a comprehensive Indigenous Knowledge System that will digitally capture, store and manage traditional knowledge, protect Indigenous intellectual property rights, facilitate intergenerational knowledge transfer, support Indigenous-led research and integrate both modern technology and Indigenous ways of knowing. The system to have the following capabilities:

1. Traditional Knowledge Repository link to oral histories, medicinal plant uses, farming techniques, traditional healing methods and ecological knowledge.
2. Cultural Heritage & Oral Traditions Archive such as stories, folklore, myths.
3. Indigenous Science and Environment Management of traditional weather forecasting, land-use management and sustainable fishing/hunting practices.
4. Traditional Medicine & Healing Practices like herbal medicines, traditional healing methods, and plant-based remedies.
5. Craftsmanship, Art and Cultural Practices such as techniques for weaving, carving, pottery, and other traditional arts.
6. Community-Driven Access and Control.
7. Indigenous Knowledge and Education Platform.
8. Traditional Governance and Decision-Making Support.
9. Security & Data Sovereignty.

Option 2: Student Administration System

You were contacted by your secondary school principal about your school's lack of proper administrative systems, making it impossible to monitor student performance, attendance, discipline records and extracurricular activities. The fragmented data management, manual record-keeping and a lack of connection between departments in schools leads to frequent issues which results in delays, errors and decision-making concerns.

Establishing a Student Administration System (SAS) helps speed up student administration, enhance record-keeping and foster better communication among students, educators and parents. This system should act as a consolidated digital platform, including technology to increase accessibility, usability and data security while providing a consistent experience for all stakeholders.

The schools currently are faced with several challenges: manual record-keeping can result in lost student information, difficulty tracking academic achievement, attendance, and disciplinary actions, limited communication among students, instructors, and parents, inefficient enrolment and scheduling management, and insufficient data for decision-making about student performance and school operations.

This systems goal is to create and run a structured SAS system that collects, stores and manages all student information which increases efficiency and minimises administrative workload. Ensure improving the communication between instructors, students and parents with a user-friendly platform. Streamline student enrolment, scheduling and academic performance monitoring. Create a centralised database that enable real-time access to student records, attendance and extracurricular activities and lastly, maintain data security and comply with student privacy standards.

The SAS system should provide the following:

1. A repository for student information.
2. A report card system and academic performance tracker.
3. Discipline and attendance monitoring and management.
4. Class timetables and Scheduling System
5. Extra curricular and student activities tracker system.
6. Communication and parental engagement platform.
7. Staff management module.
8. Library and learning resources system.
9. Data privacy and security.

Option 3: Second-hand Clothing System

With the growing trend for second-hand clothing industry buyers are seeking unique fashion clothing, affordable clothing and sustainable clothes. Small businesses and individuals are often seen buying and selling pre-owned clothing, however, they are faced with various challenges like poor advertising, inventory management and reaching out to correct consumers.

It has been noticed that small businesses and individuals who sell second-hand clothing rely mostly on informal means such as word of mouth and social media. Such means can be inefficient, resulting in limited marketing, inconsistent pricing and trouble monitoring transactions. This has a reciprocal relationship to the buyers, for instance, buyers struggle to choose the best suitable and fashionable second-hand clothes that fulfills their expectations because there is no centralised platform for browsing and purchasing products.

A Second-hand Clothing Advertisements and shopping System (SCARS) must be developed to streamline the purchasing and selling of used clothing, enhance advertising efficiency and provide a seamless shopping experience. This system must function as a digital marketplace, including contemporary technology to increase accessibility, usability and security for both consumers and sellers.

Buyers and sellers are now experiencing a number of obstacles, including a lack of centralised platforms, inconsistent pricing and difficulty in valuing things, limited exposure for sellers, trust concerns, inefficient inventory management and limited eco-friendly buying alternatives.

The project's purpose is to create and deploy a complete SCARS that will serve as a digital marketplace, improve customer experience, streamline inventory management, integrate secure payment and delivery tracking systems and promote sustainable fashion.

The system should account for the following capabilities:

1. It must ensure Seller Registration and Storefront Management
2. Proper advertising module and Second-hand Clothing Listings
3. AI-Powered Matching and Recommendation System
4. Inventory and Order Management System
5. Secure Payment & Transaction Module
6. Eco-Friendly and Sustainability Metrics
7. Integration Delivery and Logistics
8. User Reviews and Trust System
9. Security and Data Privacy

Assignment 1 Mark Allocation Sheet

After having discussed it as a group, we recommend the following mark allocation to each group member based on contribution or lack of it throughout the assignment.

Group number _____ Project manager _____

Member Student ID	Percentage Contribution	Responsibilities

Signed

Member Name	ID	Signature
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_____	_____	_____
_____	_____	_____
_____	_____	_____
_____	_____	_____

Assignment 1 Assessment Rubric

Criterion	CBOK	Beyond expectation [8.0-10.0] [16.0-20.0]	Meet expectation [5.0-7.9] [10.0-15.9]	Below Satisfactory [0.0-4.9] [0.0-9.9]	Score
Content	Abstraction And Design - V, VI, VII	The report content is well written, is complete in all aspects, and satisfies all assignment requirements.	The report content is well written, is complete in most aspects, and satisfies most assignment requirements.	The report content is inadequate, is incomplete in many aspects, and satisfies very few requirements	20
Organization	Communication - IV	The report is well organized, appropriately sectioned, uses diagrams when appropriate, important issues clearly stated	Report reasonably well documented and organized.	The report is not well organized, however lacks coherence.	10
Presentation Style	Communication - V	Document literature demonstrates sophisticated clarity, conciseness, and correctness. The report is well structured and contains appropriate references	Document literature is accomplished in terms of clarity, conciseness, and contains only a few errors. The report is adequately structured and contains appropriate references.	Document literature lacks clarity or conciseness and contains numerous errors. The report is poorly structured and references are inadequate or missing.	10
Problem Analysis: (Formulate the most appropriate solution to the open-ended problem)	Abstraction and Design - V, VI, VII	☐ Identifies information requirements and selects what is relevant from the open literature.	☐ Identifies information requirements and is a selection of the relevant from the open literature is not fully evident.	- Does not identify relevant information from open literature. - Does not identify relevant constraints and	20

		Identifies all relevant constraints and requirements and formulates an accurate description of the problem	Identifies most relevant constraints and requirements and formulates a description of the problem.	requirements and is not able to formulate a description of the problem.	
Design of solution (Formulate the most appropriate solution to the open-ended problem)	Abstraction and Design - V, VI, VII Complex computing - I, II, III V, VI	Systematically and routinely adopts a structured and professionally recognized approach to design activity. Demonstrates creativity when proposing possible solutions. Screens alternative solutions systematically. Evaluates the feasibility of several possible solutions in all relevant contexts which, as appropriate to the problem, may include: technical, suitability for implementation, economic, aesthetic, ethical, health and safety, societal, environmental, and cultural.	A structured and professionally recognized approach to design activity lacks some systematic approach. Demonstrates some aspects of creativity when proposing possible solutions. Screens alternative solutions but some systematic approaches are absent. Evaluates the feasibility of some possible solutions in some relevant contexts which, as appropriate to the problem, may include: technical, suitability for implementation, economic, aesthetic	The systematic approach to design is not fully evident. Most aspects of creativity are lacking. The solution to the open-ended problem is not fully evident. Does not show evidence of a systematic approach to screening for solutions. Evaluation of feasibility is not evident.	20

Team Work: (Operate effectively as a team member/leader)	Teamwork concepts/issues - I, IV, V	Manages own activities with honesty and integrity in an orderly manner to meet deadlines. Contributes constructively to team decision making, earns the trust and confidence of other team members. Provides leadership in a team environment by making informed decisions, keeping the team motivated, and accepting and delegating responsibility (where applicable).	Manages own activities with honesty and integrity in an orderly manner to meet deadlines most of the time. Contributes constructively to team decision making, earns the trust and confidence of other team members most of the time. Provides leadership in a team environment by making informed decisions, keeping the team motivated, and accepting and delegating responsibility most of the time applicable).	Unable to manage own activities with honesty and integrity in an orderly manner to meet deadlines most of the time. Unable to contribute constructively to team decisionmaking, earn the trust and confidence of other team members most of the time. Is not able to provide effective leadership and hence the team.	10
Feasibility of design (reflected in implementation plan)	IT Project management	The plan is well-established, manageable, appropriate, and comprehensive.	There is a logical and thoughtful plan for the manageable execution of project.	The implementation plan outlined is unclear, does not follow from design artifacts, and/or does not seem entirely manageable.	10

PART 2: Presentation

This section covers the online presentation of the project (Check Moodle for Due Date): Students will have a maximum of 7 minutes to present and 3 minutes allocated for the Q&A session. (A copy of the Recorded Presentation is required to be uploaded in the appropriate presentation dropbox).

Title of the Project	
Name of Students	
Names of Supervisors	
Name of the Panel Member	

Panel comments

The scope of the project is appropriate for a second-year undergrad level and the students should be able to complete it within 10 weeks (5-week A&D and % week Implementation)?			
Does the project satisfy the criteria for complex computing problems?			
Any regulatory approvals still required (e.g. ethics, privacy, or copyright clearances)			
Marking Criteria		CBOK	Marks
1	An appropriate flow of the requirements and UML clearly described (3 marks)	Comm.	
2	Communication skills (3 marks)	Comm.	
3	Students answered the questions well (4 marks)	Comm.	
4	The project satisfies the criteria for complex computing – level M problem (15 marks)	Com. Comp.	
Total (25)			
Date			
Signature of the Panel Member			

The End